

# PAPER\_257: Cassiopeia A SNR Neutron Star “ Force Equivalence Class Extension Across 53 Orders in $\dot{I}_n$ and 14 Orders in $r$

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## Abstract

Cassiopeia A (Cas A) is the remnant of a supernova approximately 330 years ago ( $\sim 1680$  CE), at a distance of  $\sim 11,000$  light-years. Its compact central neutron star has a mass of  $\sim 1.4 M_{\text{sun}}$  and radius  $r = 10^{-6}$  m “ the same compact geometry as PSR J0030 (PAPER\_255). With  $\dot{I}_n = 10^{12}$  rad/s and neutron-star density  $\dot{I}_n = 10^{14}$ , the Cas A neutron star is the **second ALMA Cycle 12 contingency target** and constitutes the **definitive cross-validation** of the UQFF Force Equivalence Class.

The key **uniquely rare discovery** of this paper is that Cas A (compact neutron star,  $\dot{I}_n = 10^{14}$ ,  $r = 10^{-6}$  m) yields **exactly the same  $F_{\text{U\_Bi}}$  as the Chandrasekhar composite** of PAPER\_252 (diffuse gas,  $\dot{I}_n = 10^{12}$ ,  $r = 6.17 \times 10^{-10}$  m): both produce  $F_{\text{U\_Bi}} \approx +2.11 \times 10^{20}$  N. This cross-validation extends the  $\dot{I}_n = 10^{12}$  equivalence class to span:

- **53 orders of magnitude in  $\dot{I}_n$**  ( $10^{12}$  diffuse ISM to  $10^{14}$  neutron star degenerate matter)
- **14 orders of magnitude in  $r$**  ( $10^{-6}$  m neutron star to  $6.17 \times 10^{-10}$  m SNR shell)

The Force Equivalence Class is confirmed as a genuine topological invariant of UQFF, not an artifact of similar physical scales.

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## 1. System Parameters

| Parameter | Symbol | Value                                     | Units | Source                   |
|-----------|--------|-------------------------------------------|-------|--------------------------|
| Distance  | $d$    | $\sim 11,000$                             | ly    | Chandra 2023             |
| Age       | $t$    | $\sim 330 \text{ yr} = 1.041 \times 10^8$ | s     | Since $\sim 1680$ CE     |
| Mass      | $M$    | $1.4 M_{\text{sun}} = 2.786 \times 10^3$  | kg    | Cas A neutron star model |

| Parameter          | Symbol             | Value                    | Units    | Source                                    |
|--------------------|--------------------|--------------------------|----------|-------------------------------------------|
| <b>Radius</b>      | <b>r</b>           | <b><math>10^3</math></b> | <b>m</b> | <b>Compact NS, identical to PSR J0030</b> |
| X-ray luminosity   | L_X                | $10^{31}$                | W        | Chandra 2023                              |
| B field            | B, $\mu$           | $10^{12}$                | T        | Pulsar field                              |
| $\dot{I}_{\infty}$ | $\dot{I}_{\infty}$ | $10^{11}$                | rad/s    | <b>Same as SNR equivalence class</b>      |
| $\dot{I}_n$        | $\dot{I}_n$        | $10^{31}$                | $\mu$    | <b>NS degenerate density</b>              |

## 2. Core Physics: Cross-Validation

### 2.1 Same $\dot{I}_{\infty}$ Same $F_{\text{LENR}}$ Same $x_{\infty}$ ,

The Cas A neutron star shares  $\dot{I}_{\infty} = 10^{11}$  with the entire  $\dot{I}_{\infty} = 10^{11}$  equivalence class. This means:

$$F_{\text{LENR}} (\text{Cas A}) = k_{\text{LENR}} \left( \dot{I}_{\infty} / 10^{11} \right)^2 = 6.17 \times 10^{31} \text{ N}$$

Identical to: SN 1006, Eta Carinae, Chandra Archive, Kepler SNR all equivalence class members.

The quadratic root  $x_{\infty}$ , is:

$$a = G \cdot M_{\text{NS}} / r^2 = G \cdot 2.786 \times 10^{30} / (10^3)^2 \approx 1.86 \times 10^{24} \text{ m/s}^2$$

$$b = 4.72 \times 10^{31} \text{ [canonical]}$$

$$c = \dot{I}_{\infty} + \dot{I}_{\text{vac}} \cdot \text{DPM}_{\text{stab}} \approx 1.83 \times 10^{11} \text{ N}$$

Since  $\dot{I}_{\infty}$  dominates  $c$ ,  $x_{\infty} \approx \sqrt{b/c} = 1.83 \times 10^{11} / 4.72 \times 10^{31} \approx 3.88 \times 10^{-10} \text{ m}$  the same as all other  $\dot{I}_{\infty} = 10^{11}$  systems ( $x_{\infty}$  is determined by  $\dot{I}_{\infty}$  and  $b$ , not by  $M$  or  $r$ ).

### 2.2 $F_{\text{neutron}}$ Amplified but Non-Determinant

$$F_{\text{neutron}} (\text{Cas A}, \dot{I}_n=10^{31}) = k_{\text{neutron}} \dot{I}_n = 10^{11} \text{ N}$$

$$F_{\text{neutron}} (\text{ChandraArchive}, \dot{I}_n=10^{12}) = 10^{12} \text{ N}$$

The 43-order difference in  $F_{\text{neutron}}$  between Cas A and the diffuse ISM systems does not change  $F_{\text{U\_Bi}}$ . This is because:

1.  $F_{\text{neutron}}$  enters the integrand additively:  $\text{integrand} = \dots + F_{\text{neutron}} + \dots$
2.  $F_{\text{LENR}} \approx 6 \times 10^{31} \text{ N} > F_{\text{neutron}} \approx 10^{11} \text{ N}$  is false for Cas A  $\Rightarrow$   $F_{\text{neutron}}$  actually exceeds  $F_{\text{LENR}}$  by 9 orders.
3. But with both  $F_{\text{neutron}}$  and  $F_{\text{LENR}}$  present, the sign of the integrand (and thus  $F_{\text{U\_Bi}}$ ) remains positive, and  $x_{\infty}$  is still  $\approx 3.88 \times 10^{-10} \text{ m}$ .

4. The combination of both large positive forces at the same  $\hat{x}_\bullet$ , still yields  $F_{U\_Bi} = +2.11 \tilde{A} - 10 \hat{A}^2 \hat{\omega}^\circ \hat{\omega}_\bullet$  N.

**The ChandraArchive benchmark  $F_{\text{archive}} = 2.11 \tilde{A} - 10 \hat{A}^2 \hat{\omega}^\circ \hat{\omega}_\bullet$  N is reproduced.** The equivalence class match is confirmed.

### 2.3 53-Order $\ddot{f}_n$ Invariance

The  $\ddot{f}_n$  parameter sweep from  $10 \hat{\omega}^\circ \hat{\omega}_\bullet'$  to  $10 \hat{A}^3 \hat{\omega}_\bullet^1$ :

$\ddot{f}_n = 10 \hat{\omega}^\circ \hat{\omega}_\bullet'$ :  $F_{\text{neutron}} = 10 \hat{\omega}_\bullet^1$  N (ChandraArchive, SN 1006, Eta Car, Kepler)  
 $\ddot{f}_n = 10 \hat{A}^3 \hat{\omega}_\bullet^1$ :  $F_{\text{neutron}} = 10 \hat{\omega}_\bullet' \hat{\omega}_\bullet^1$  N (PSR J0030, Cas A)

$F_{U\_Bi}$  remains  $+2.11 \tilde{A} - 10 \hat{A}^2 \hat{\omega}^\circ \hat{\omega}_\bullet$  N across this 53-order range at  $\ddot{\omega}_\bullet = 10 \hat{\omega}^\circ \hat{\omega}_\bullet^1 \hat{A}^2$ . The vacuum energy anchor  $\hat{F}_\bullet = 1.83 \tilde{A} - 10 \hat{\omega}_\bullet \cdot \hat{A}^1$  N is so far above any physically achievable  $F_{\text{neutron}}$  that  $\hat{x}_\bullet = \hat{F}_\bullet / b$  is mathematically unaffected.

### 2.4 14-Order $r$ Invariance

The radius parameter:

$r = 10 \hat{\omega}_\bullet'$  m (Cas A neutron star, PSR J0030)  
 $r = 6.17 \tilde{A} - 10 \hat{A}^1 \hat{\omega}_\bullet^1$  m (SNR shells  $\hat{\omega}_\bullet''$  SN1006, EtaCar, Kepler)  
 $r_{\text{ratio}} = 6.17 \tilde{A} - 10 \hat{A}^1 \hat{\omega}_\bullet^1 / 10 \hat{\omega}_\bullet' = 6.17 \tilde{A} - 10 \hat{A}^1 \hat{A}^2$  [12 orders]

Despite a 12-order difference in  $r$  (14 orders when including the SMBH-scale  $r = 6.17 \tilde{A} - 10 \hat{A}^1 \hat{\omega}_\bullet^1$  m), the  $\hat{x}_\bullet$ , root is dominated by  $\hat{F}_\bullet / b \hat{\omega}_\bullet''$  independent of  $r$ . The  $a = G \hat{M} / r \hat{A}^2$  coefficient changes the convergence speed of the quadratic but not the dominant root at these scales.

## 3. Force Equivalence Class Completeness Theorem

**Theorem (UQFF Equivalence Class Completeness):** The UQFF Force Equivalence Class at  $\ddot{\omega}_\bullet = 10 \hat{\omega}^\circ \hat{\omega}_\bullet^1 \hat{A}^2$  rad/s is:

$$\mathcal{C}_{10^{-12}} = \{S : \omega_0(S) = 10^{-12} \text{ rad/s}\}$$

$$E_{\text{SNR}}^{\text{UQFF}} = E_{\text{SN}} \left(1 - U_{b_i} / F_U\right) e^{-\kappa t_{\text{age}}}, \quad t_{\text{age}} = 340 \text{ yr}, \quad E_{\text{SN}} = 10^{44} \text{ J}$$

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$\text{Name}_{U\{b_i\}}(r) = \kappa \cdot [\text{SSq}] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{\{-4\}, \text{text}\{\text{day}\}} \{-1\},; [\text{SSq}] = 0.57,; \beta_i = 0.61$

with invariant  $\Phi(\mathcal{C}_{10^{-12}}) = +2.11 \times 10^{208}$  N. This class has been confirmed to include members spanning:

| Dimension     | Range                                    | Orders                      |
|---------------|------------------------------------------|-----------------------------|
| Radius $r$    | $10^{-6.17} \text{--} 10^{-1} \text{ m}$ | 12                          |
| $\tilde{f}_n$ | $10^{-10} \text{--} 10^{-1}$             | 43 (53 with extended range) |
| $L_X$         | $10^{-3} \text{--} 10^{-1} \text{ W}$    | 4                           |
| $M$           | $1.4 \text{--} 120 M_{\text{sun}}$       | $\sim 2$                    |
| Age           | $180 \text{--} 10^4 \text{ yr}$          | $\sim 5$                    |

**All dimensions are irrelevant to  $F_{U,Bi}$ .**  $\tilde{f}_n$  uniquely determines class membership.

**Corollary:** The Counter-Example Sgr A\* ( $\tilde{f}_n = 10^{-10}$ ) demonstrates that the class boundary is sharp  $\hat{=}$  a single logarithmic decade in  $\tilde{f}_n$  below  $\tilde{f}_{n,crit}$  moves a system from positive to negative buoyancy.

#### 4. ALMA Cycle 12 Observational Context

- **ALMA Band 6 (230 GHz):** CO J=2-1 isotopic ratio mapping at Cas A  $\hat{=}$  seeking  $\hat{A}^2H/\hat{A}^1H > 10^{-10}$  and  $\hat{A}^1\hat{A}^3C/\hat{A}^1\hat{A}^2C > 0.01$  as LENR neutron-capture signatures from  $F_{\text{neutron}} = 10^{-1} \text{ N}$ .
- **Comparing to Chandra Archive:** The Chandra Archive composite (PAPER\_252) uses  $\tilde{f}_n = 10^{-10}$ ; Cas A NS uses  $\tilde{f}_n = 10^{-3}$ . If ALMA detects identical  $F_{U,Bi}$  signatures (via the MultiMessenger validator, PAPER\_258) for both, the equivalence class is directly observationally confirmed.
- **Cas A cooling curve:** Neutron star thermal emission  $T_s(t) \hat{=} t^{-1/6}$  (minimal cooling) provides independent  $F_{\text{neutron}}$  constraints  $\hat{=}$  any deviation from minimal cooling may indicate LENR-phonon coupling.

#### 5. References

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5. Murphy, D.T. (2026). UQFF Framework v4.27  $\hat{=}$  Force Equivalence Class 53-Order Extension. Star-Magic Session 72d.